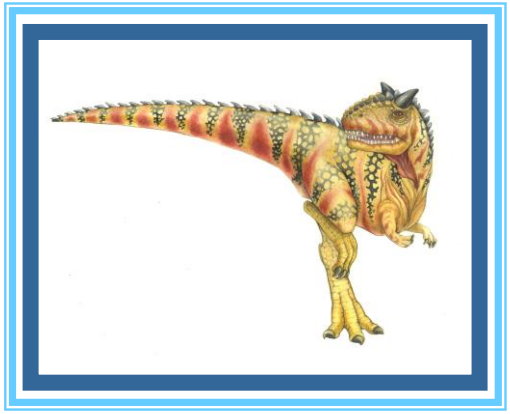
OS Lab Introduction

Chijin Zhou
East China Normal University

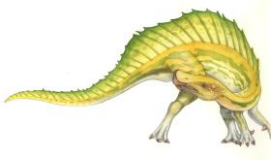




分数细节

- Lab1 基础Linux使用 （15%）
- Lab2 PintOS环境搭建（15%）
- Lab3 线程管理 （35%）
  - 上台报告15分
  - 实验报告20分
- Lab4 实现系统调用（35%）
  - 上台报告20分
  - 实验报告15分

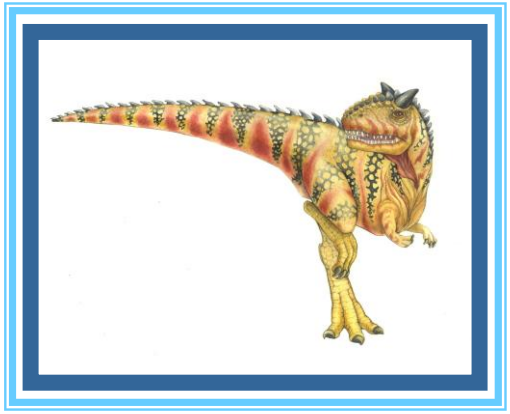
- 注：没加微信群的加微信群，没加学习通的加学习通



Using Linux and Lab Setup

OS lab 1

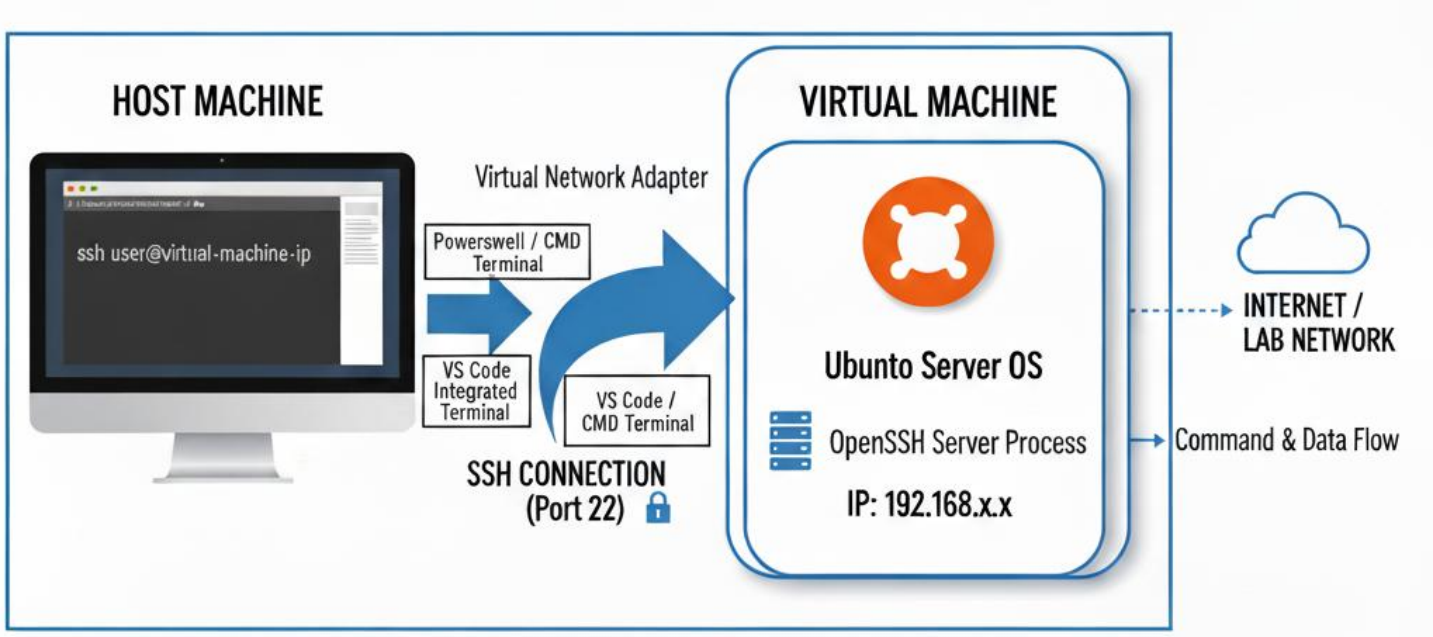
Chijin Zhou
East China Normal University





Lab1 Overview

1. 构建 Ubuntu 虚拟机（如用WSL可忽略）
2. 使用 ssh 连接到 Ubuntu 虚拟机
3. 在 Ubuntu 虚拟机上进行开发







Download VirtualBox (or VMWare)

- https://www.virtualbox.org/wiki/Downloads

Download VirtualBox

The VirtualBox Extension Pack is available for personal and educational use on this page under the PUEL license. The VirtualBox Extension Pack is also available under commercial or enterprise terms. By downloading, you agree to the terms and conditions of the respective license.

VirtualBox Platform Packages

VirtualBox 7.2.2 platform packages

- Windows hosts
- macOS / Intel hosts
- macOS / Apple Silicon hosts
- Linux distributions
- Solaris hosts
- Solaris 11 IPS hosts

Platform packages are released under the terms of the GPL version 3

VirtualBox Extension Pack

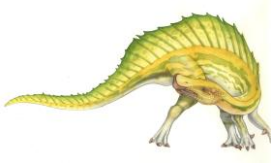
VirtualBox 7.2.2 Extension Pack

This VirtualBox Extension Pack Personal Use and Educational License governs your access to and use of the VirtualBox Extension Pack. It does not apply to the VirtualBox base package and/or its source code, which are licensed under version 3 of the GNU General Public License "GPL").

See our FAQ for answers to common questions.

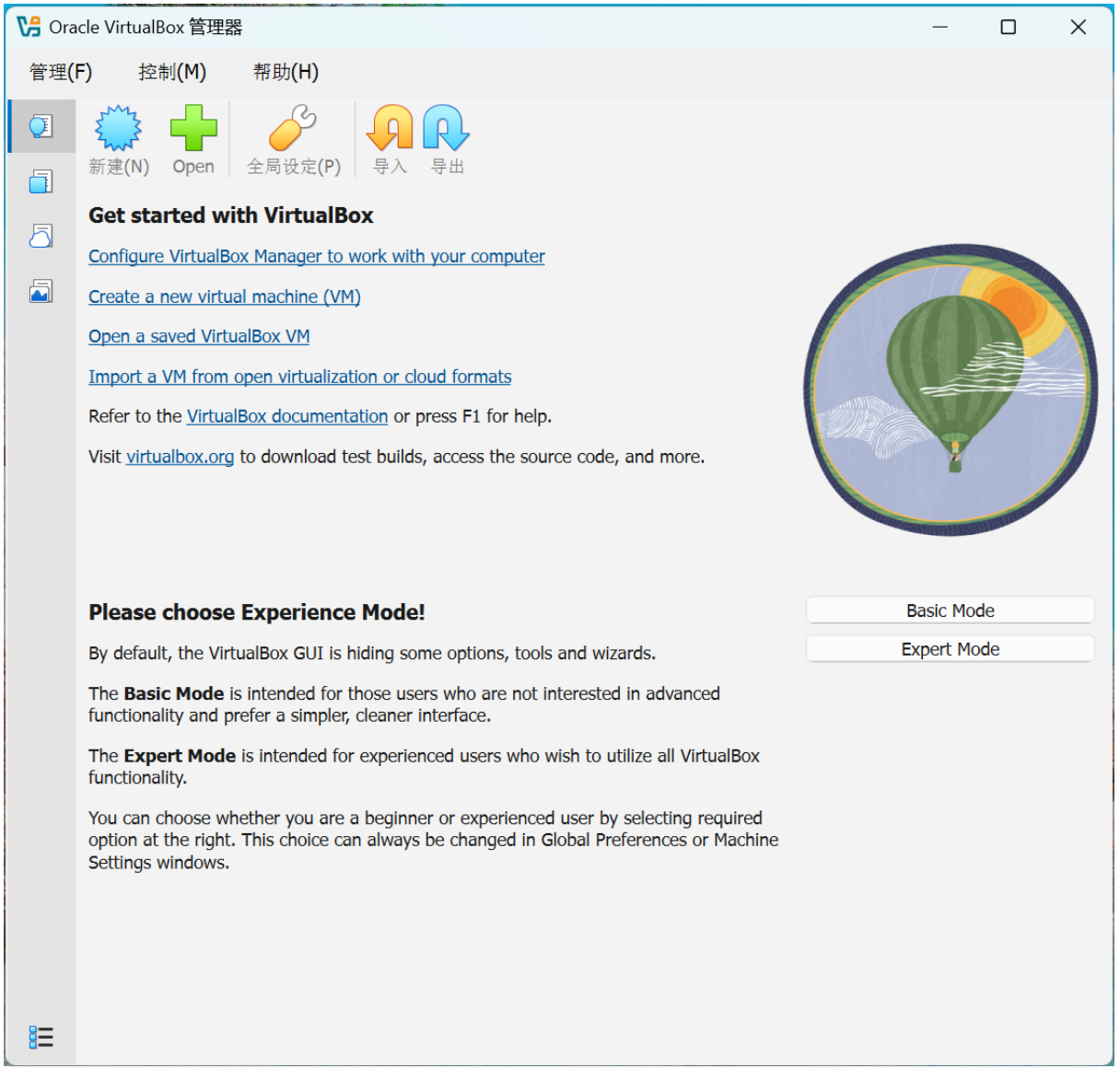
VirtualBox Extension Pack Personal Use and Educational License

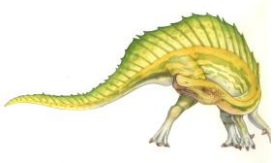
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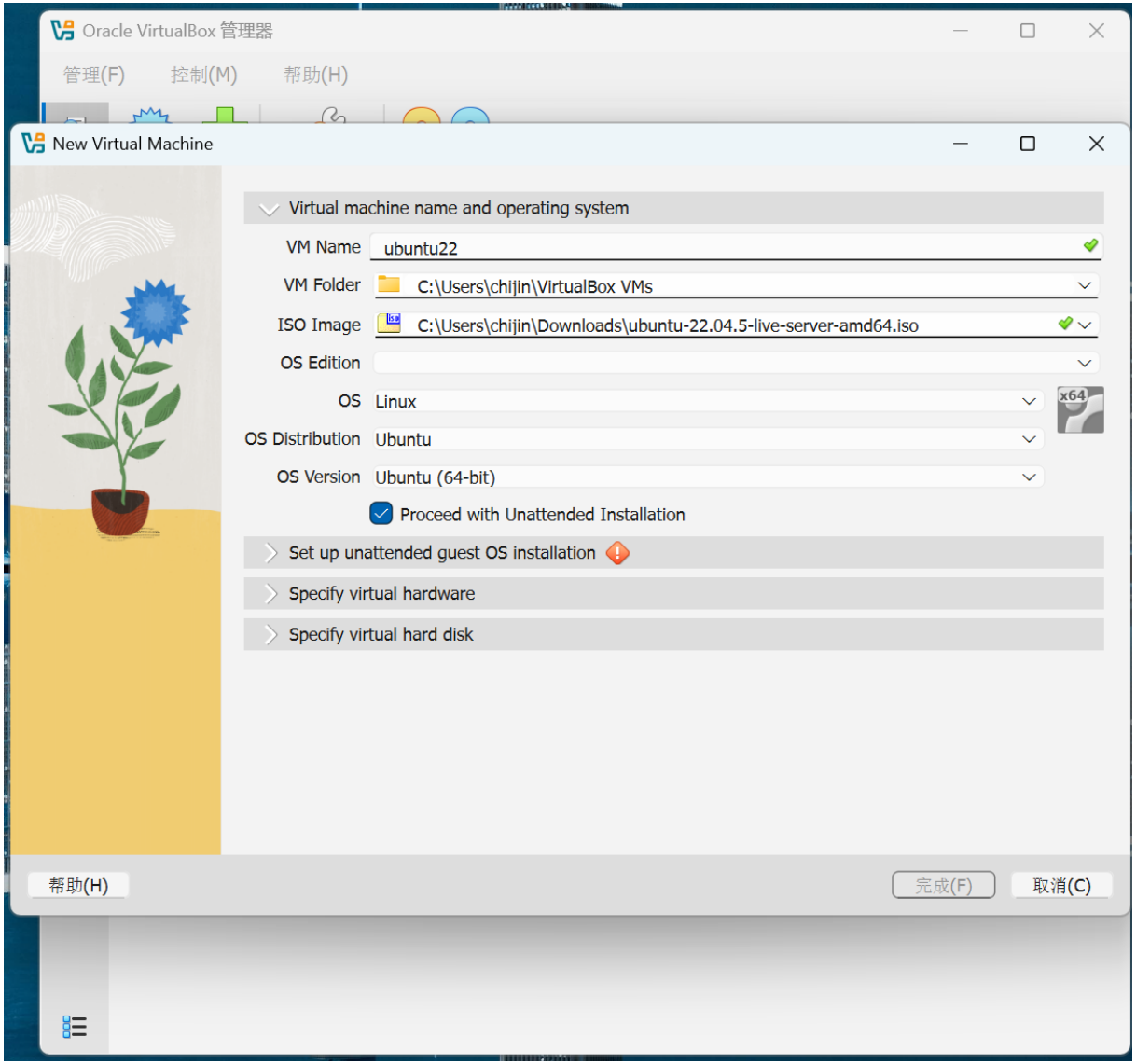
After Installing VirtualBox

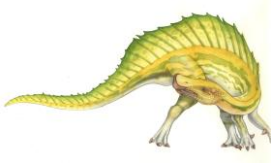




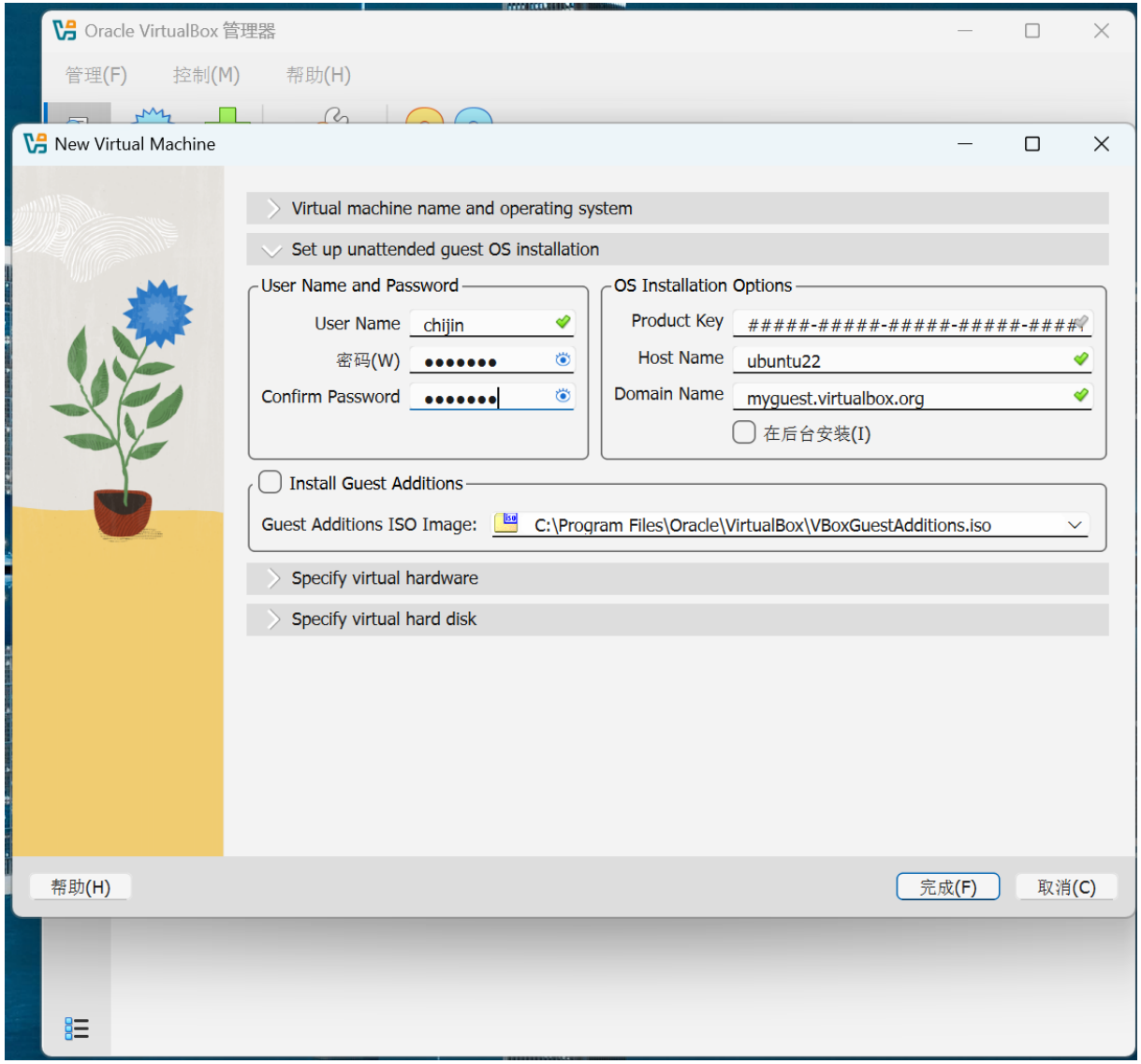


Create an Ubuntu VM





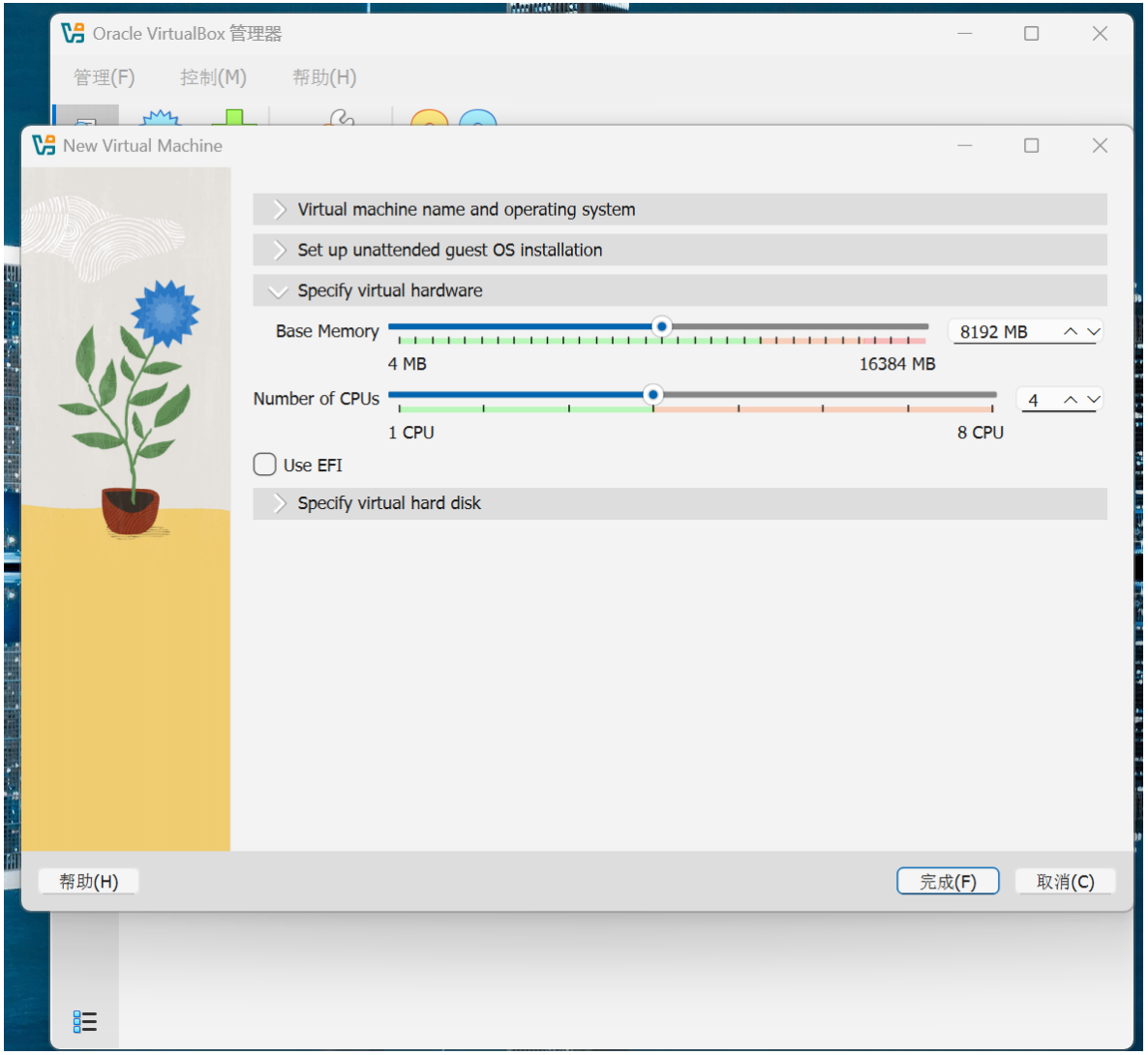
Create an Ubuntu VM



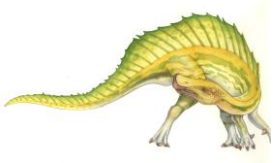
为了后续改分需要，请把用户名设为你自己的真实名字的拼音



Create an Ubuntu VM

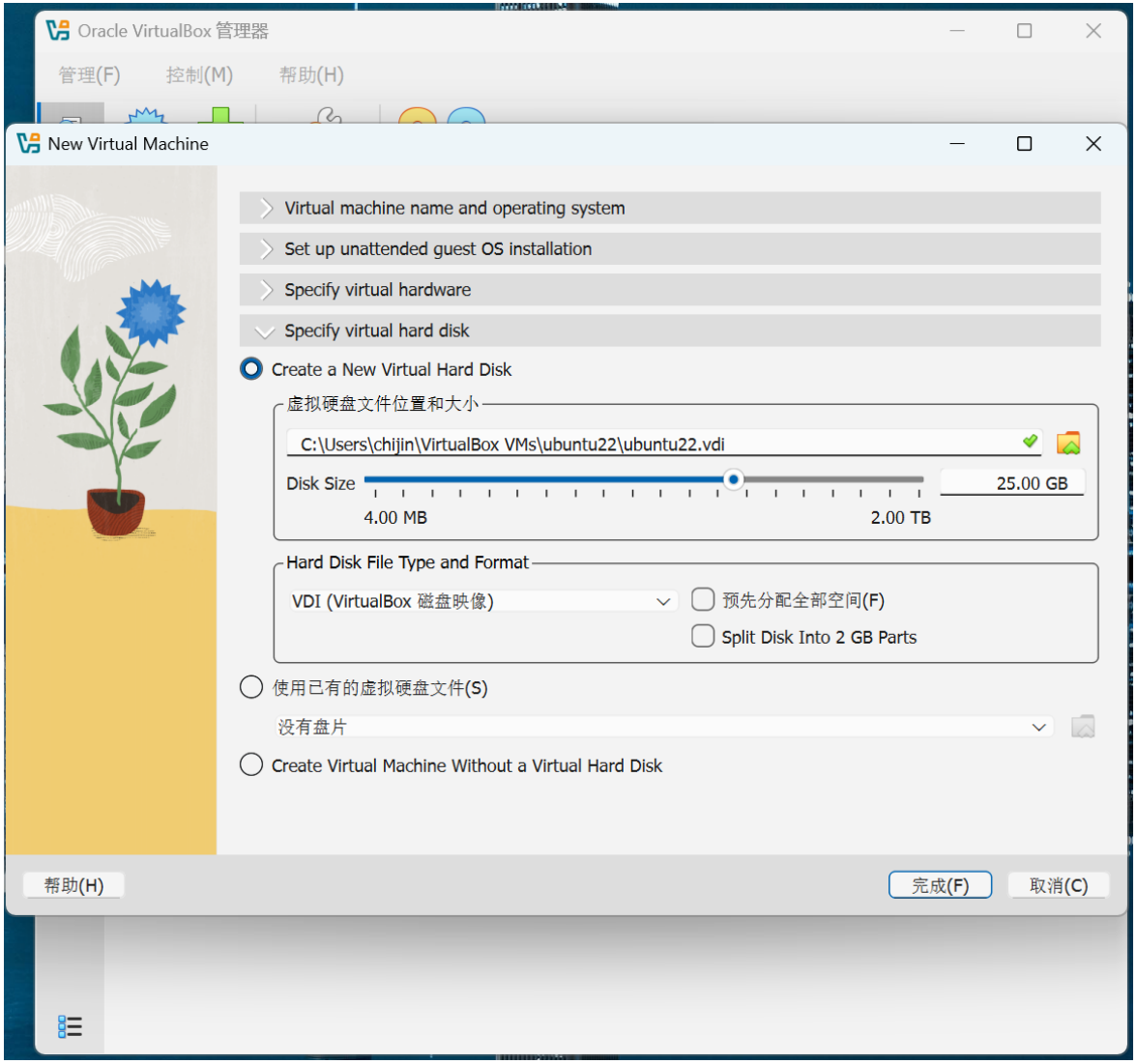


建议不少于4核、8GB内存





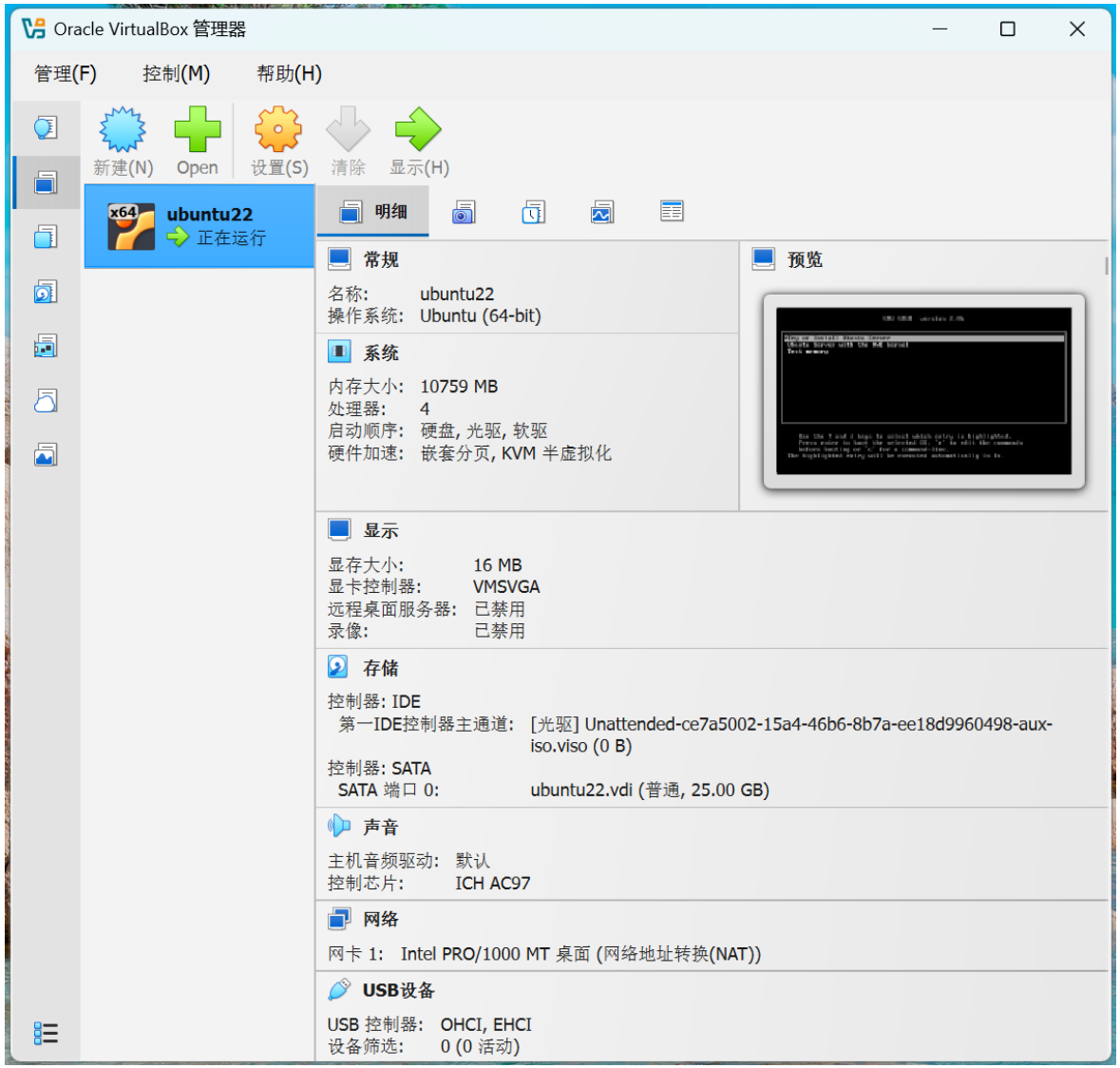
Create an Ubuntu VM





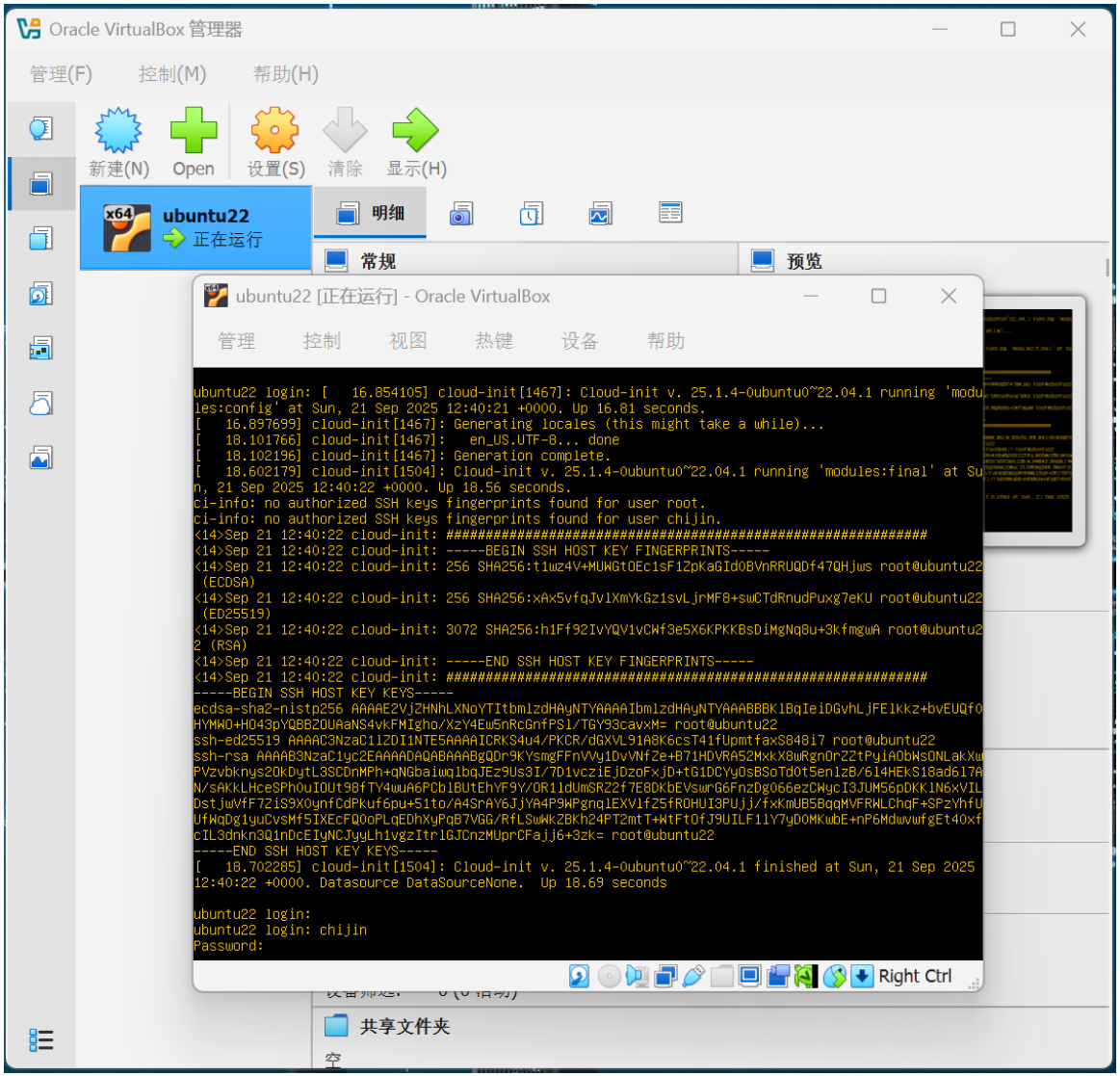


Spawn an Ubuntu VM





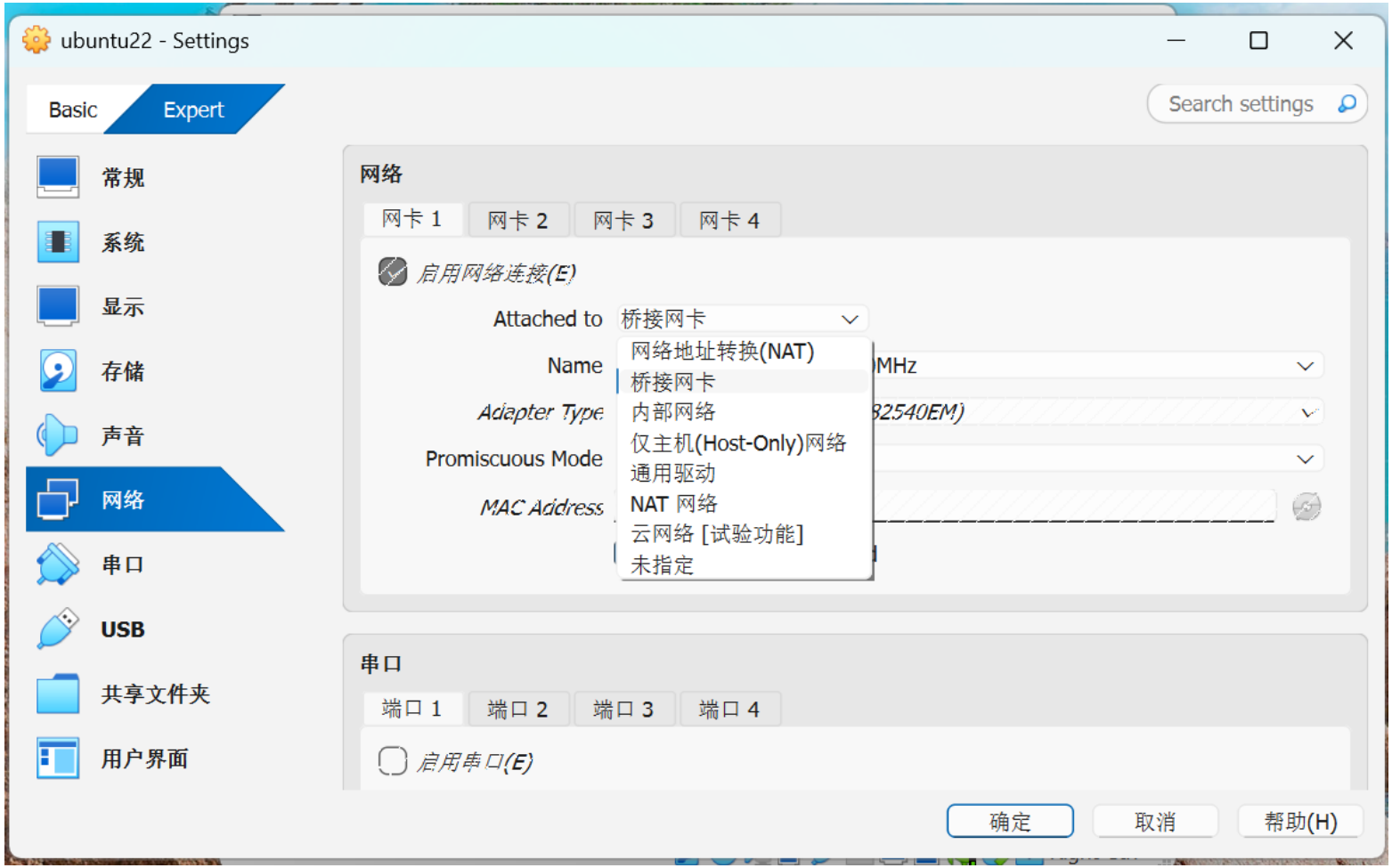
After a Long Setup…



用户名和密码是之前创建VM时设置的



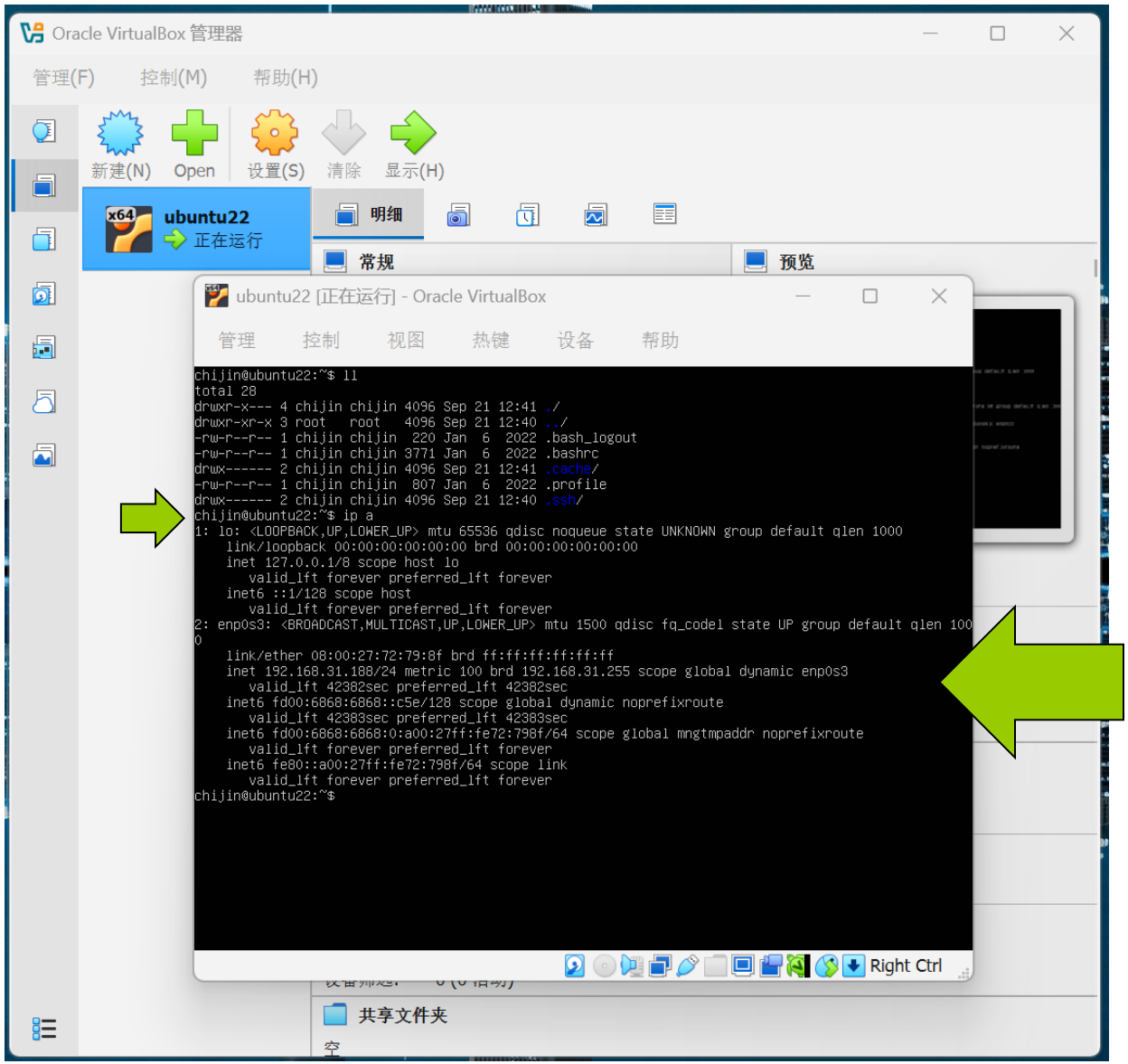
Using Bridge Adapter



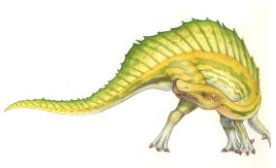




Using “ip a” to check ip



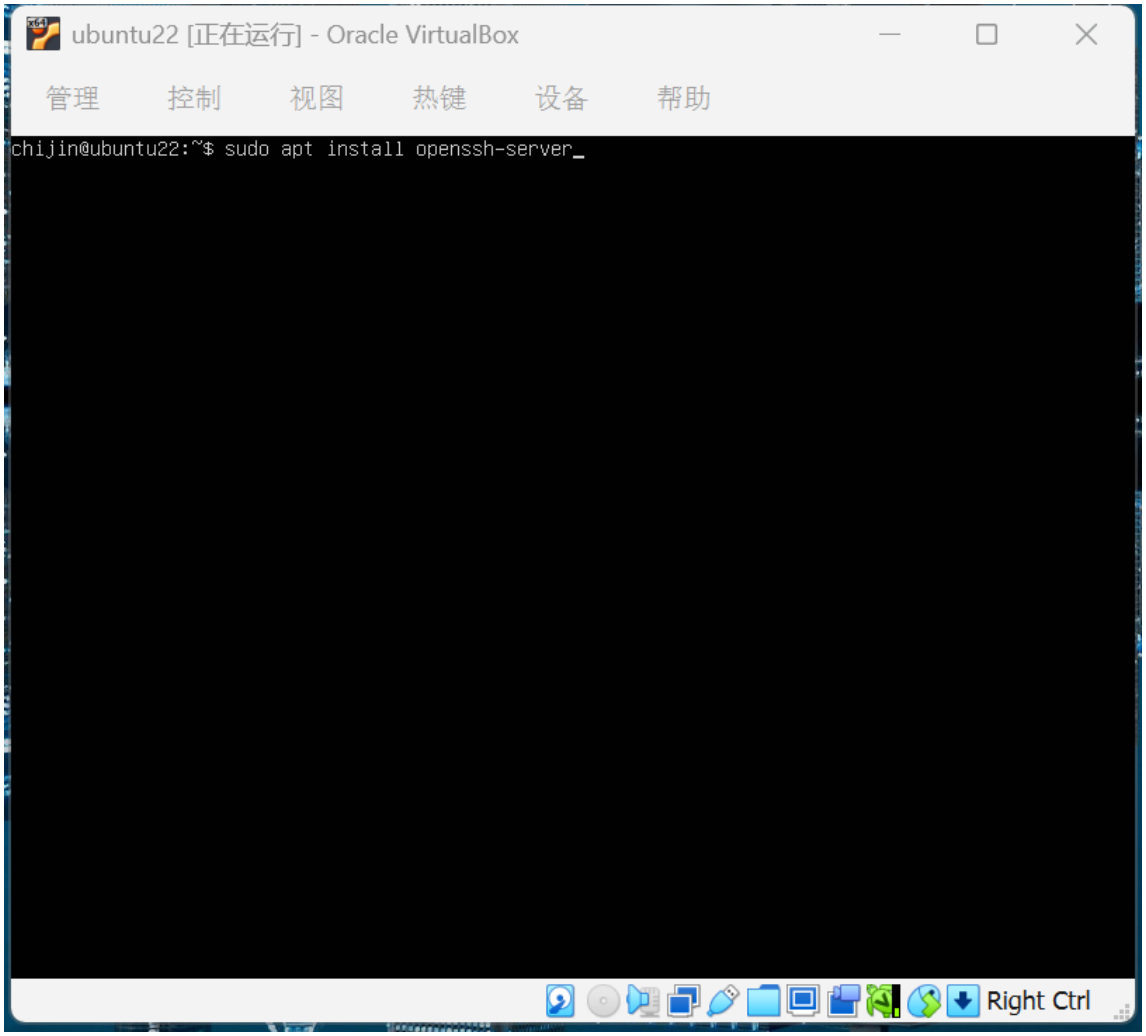
IP: 192.168.31.188

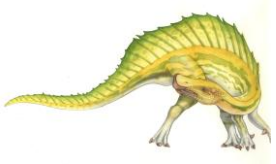




Install openssh-server

sudo apt install openssh-server



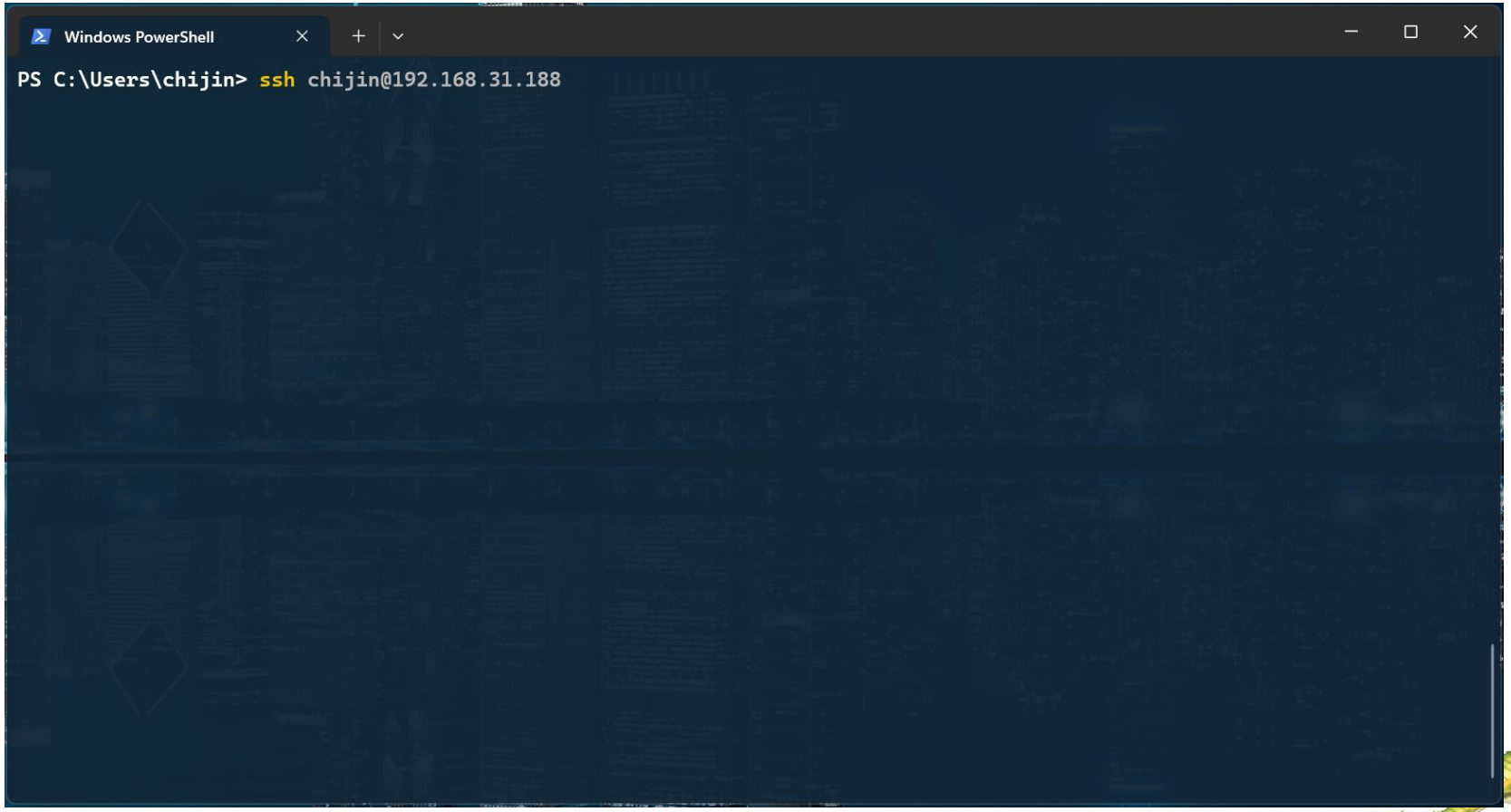


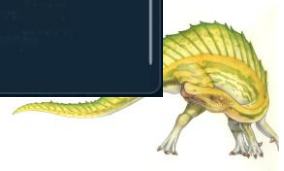


Using “ssh” to connect to the VM

In your host machine:

- open PowerShell (for Windows) or Terminal (for MacOS)
- “**ssh username@ip**”







作业1.1 成功 ssh 的截图（5分）

- 上传从你的host端主机 ssh 到 ubuntu 虚拟机的截图
- 注意你的虚拟机或用户的名称，要与你本人真实名称对应，方便批改

```
chijin@ubuntu22: ~
 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Sep 21 01:56:23 PM UTC 2025

  System load:             0.1
  Usage of /:              13.3% of 24.44GB
  Memory usage:            3%
  Swap usage:              0%
  Processes:               130
  Users logged in:         1
  IPv4 address for enp0s3: 192.168.31.188
  IPv6 address for enp0s3: fd00:6868:6868::c5e
  IPv6 address for enp0s3: fd00:6868:6868:0:a00:27ff:fe72:798f

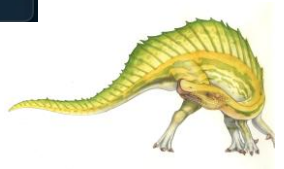

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.3 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

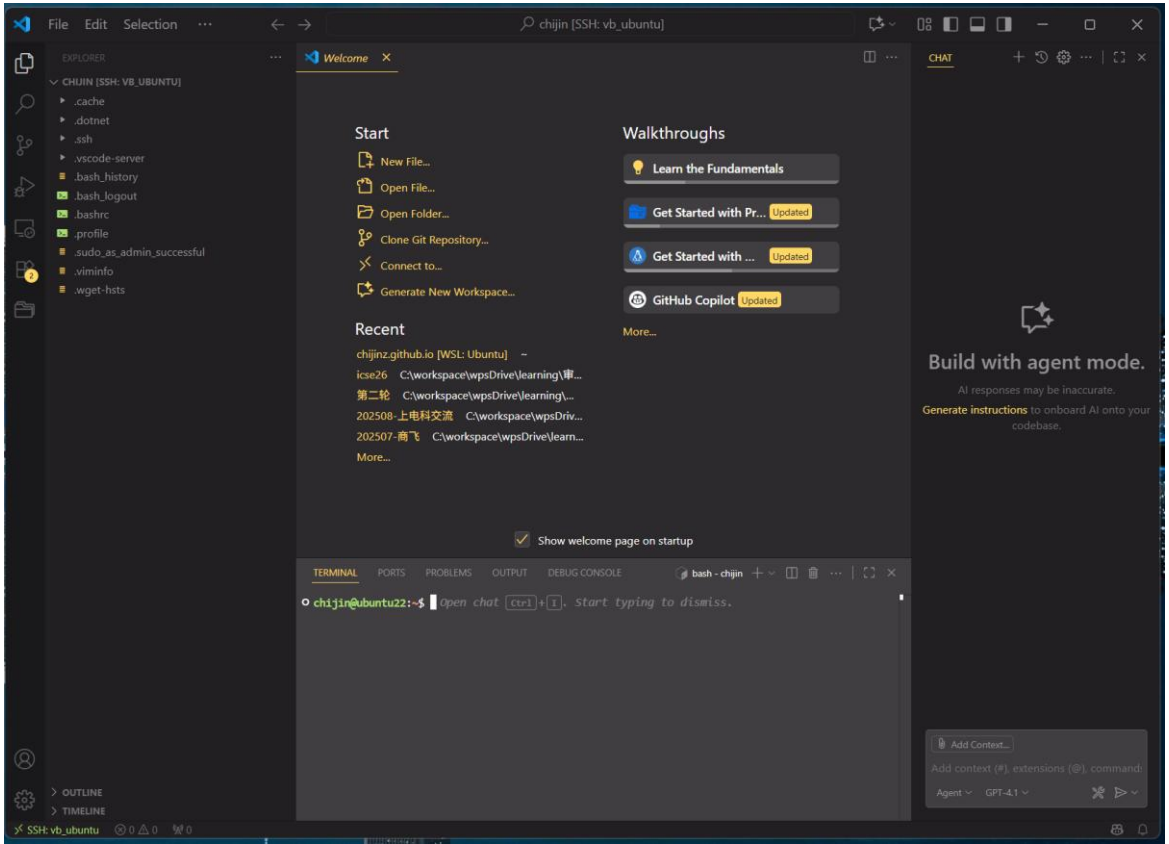

Last login: Sun Sep 21 13:55:41 2025 from 192.168.31.47
chijin@ubuntu22:~$
```

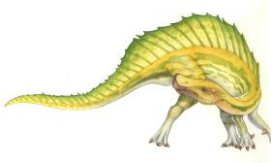




作业1.2 使用VS Code连接VM（5分）

- 使用大语言模型询问以下的功能如何操作
  - 下载VS Code，使用VS Code的 ssh 方式连接到 VM
  - 配置 ssh 相关信息（使用ssh key免密登录；写.ssh/config文件）
- 上传成功连接的截图

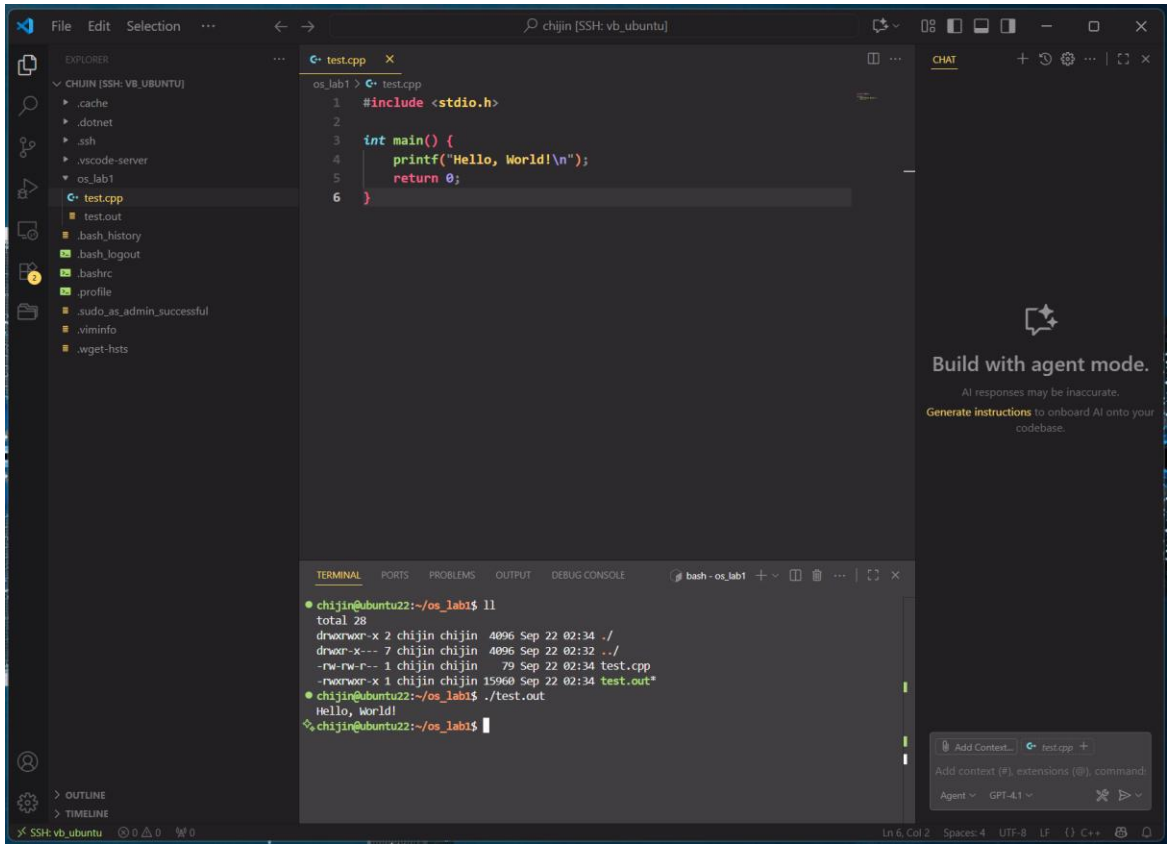






作业1.3 在VM中跑一个程序（5分）

- 使用大语言模型询问以下的功能如何操作
  - 如何在Ubuntu下编译软件 C/C++ 程序
  - 如何在Ubuntu下跑一个程序
- 上传成功跑hello world程序的截图





End of OS lab 1

